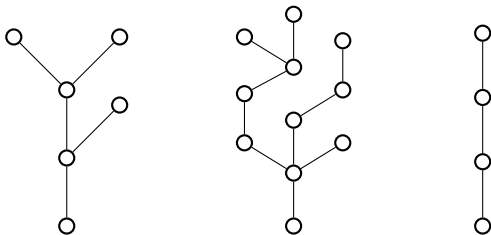


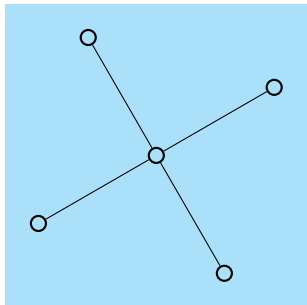
A Forest Building Process on Simple Graphs

Aurora Hiveley
Adv. Kristin Heysse

Department of Mathematics, Statistics, and Computer Science
Macalester College



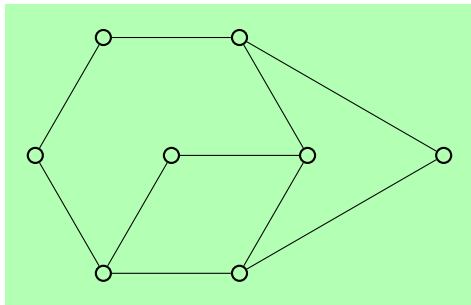
Graph Theory



$$|V| = 5, |E| = 4$$

Connected ✓

Tree ✓

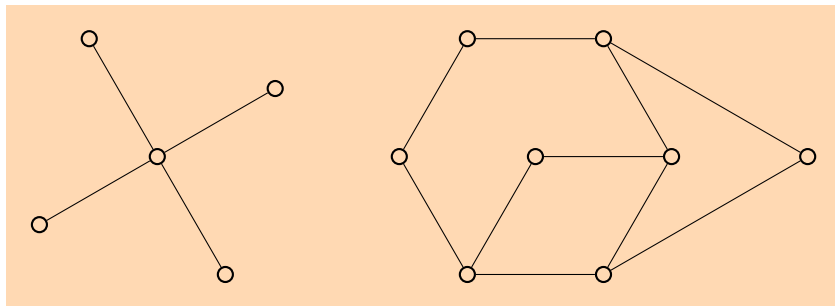


$$|V| = 8, |E| = 10$$

Connected ✓

Tree X

Graph Theory



$$|V| = 13, |E| = 14$$

Connected X

Components = 2

$$\pi = 1342$$

$$\pi^{-1} = 1423$$

Permutations & Combinations

$$\pi = 1342$$

Peaks: $\{4\}$

$$\pi^{-1} = 1423$$

Peaks: $\{4\}$

Permutations & Combinations

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Peaks: $\{4\}$



$$4! = 24$$

Permutations & Combinations

$$\pi = 1342$$

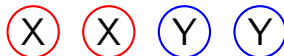
Peaks: $\{4\}$

$$\pi^{-1} = 1423$$

Peaks: $\{4\}$



$$4! = 24$$



$$\binom{4}{2} = 6$$

Forest Building

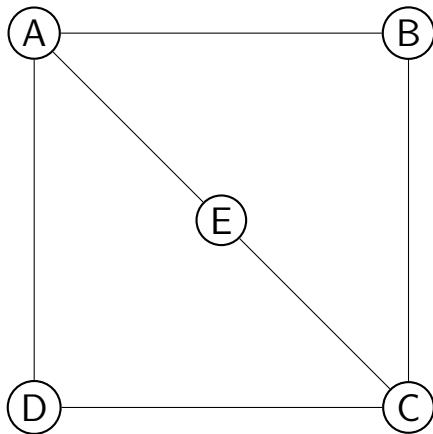
A **forest** is a graph whose components are all trees.

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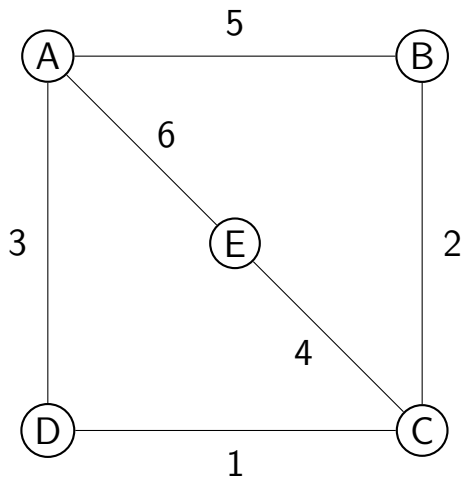
Rules

- 1.) Randomly permute the edges of G (elements of E)
- 2.) Create an empty graph from V
- 3.) Add the edges of E to the empty graph according to the random ordering *if and only if* the edge is incident to at least one isolated vertex

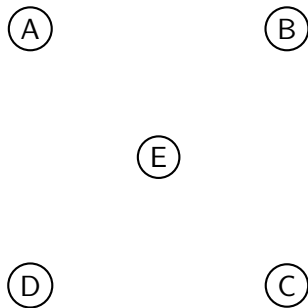
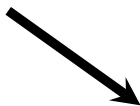
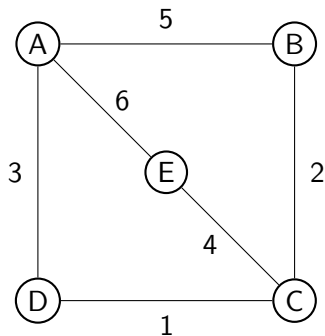
Forest Building



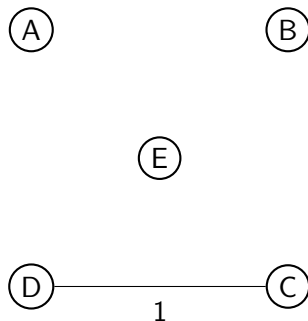
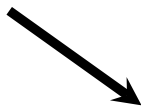
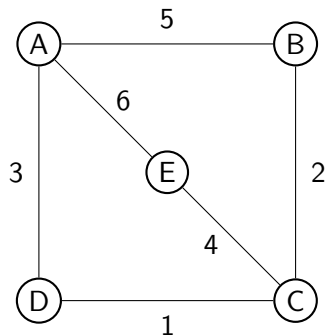
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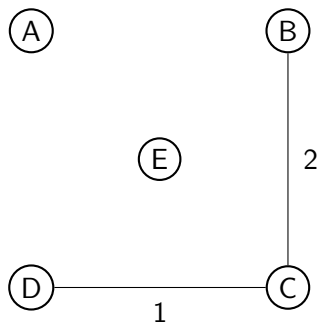
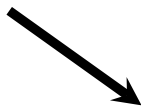
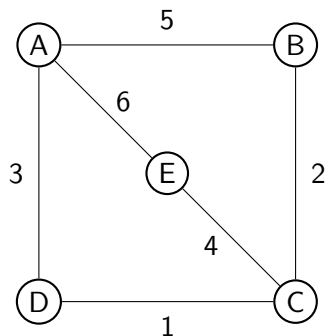
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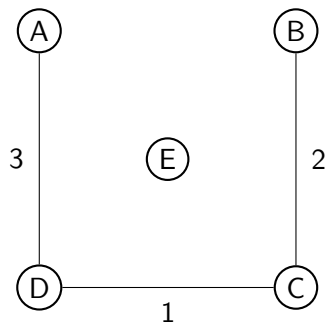
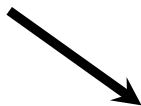
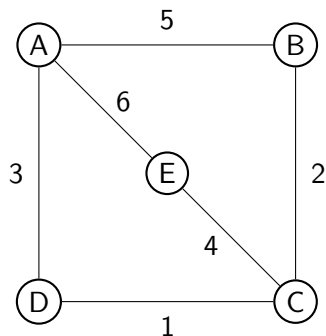
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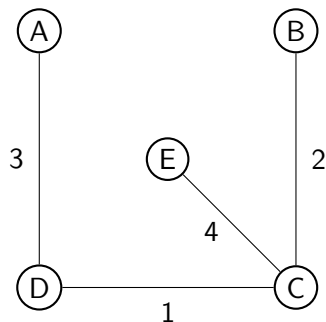
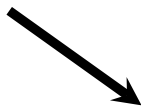
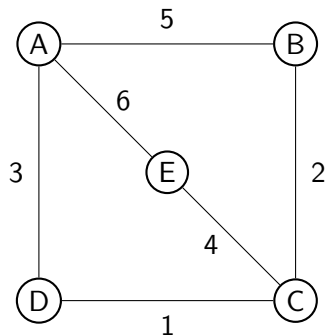
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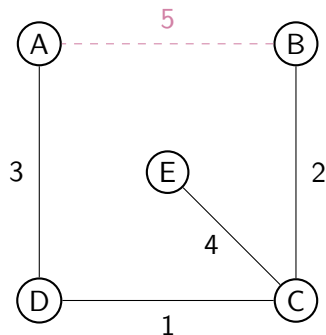
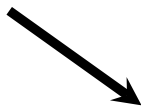
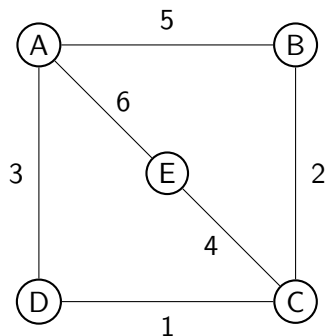
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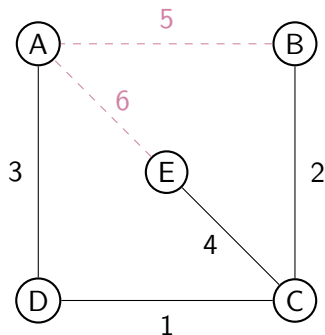
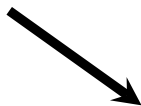
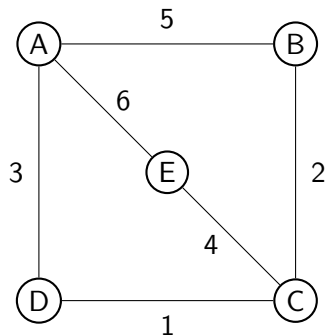
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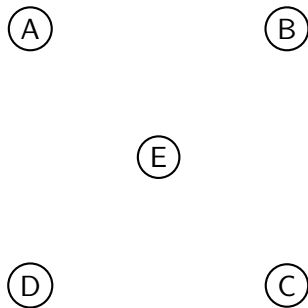
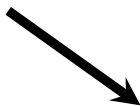
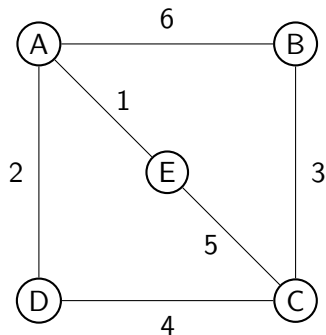
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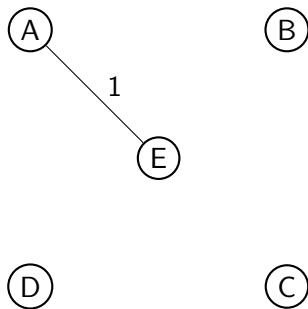
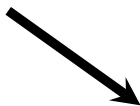
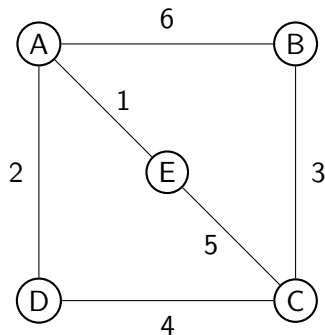
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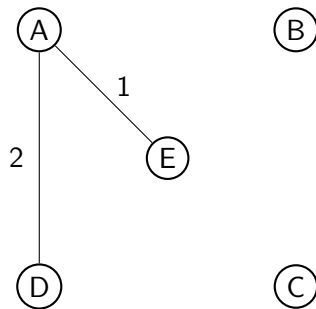
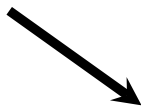
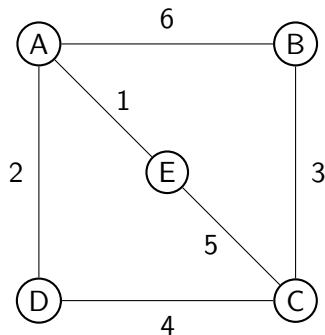
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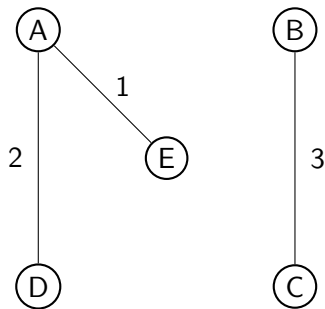
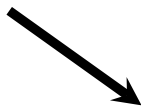
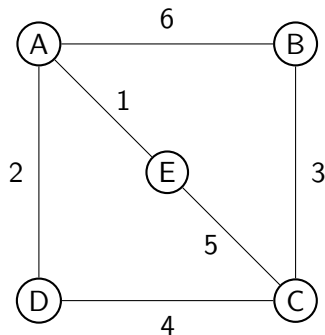
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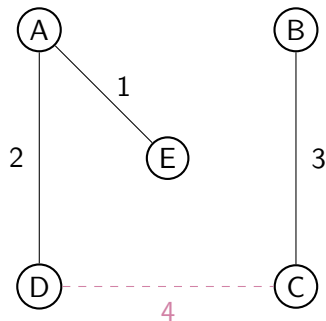
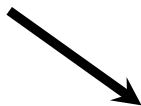
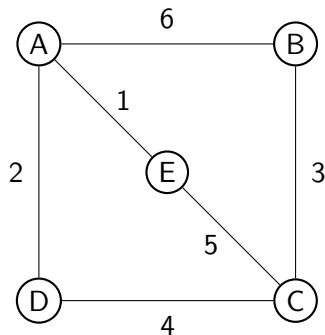
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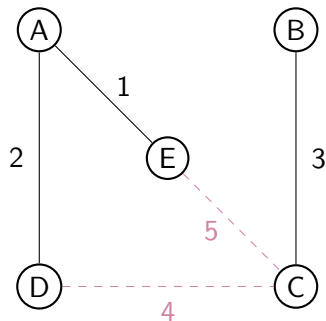
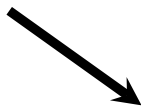
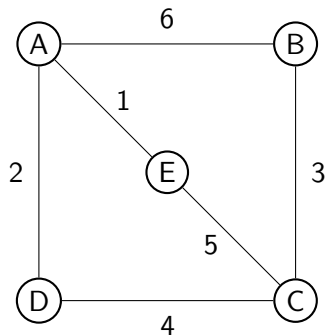
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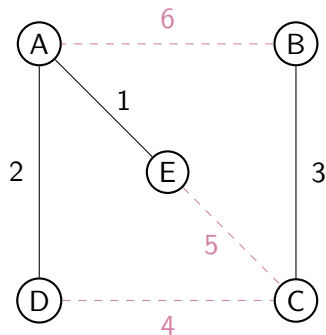
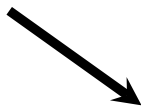
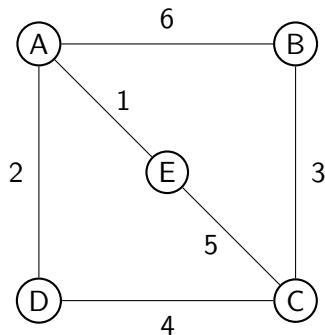
Forest Building



Forest Building



Forest Building



The Problem

- Comes from edge flips

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- Well understood: complete graphs, complete bipartite, cycles

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- Simple graph families producing at most 2 components (e.g. paws)
- Formulas for one component for some families

Questions of Interest

- Likelihood of k components?

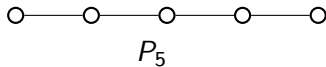
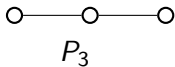
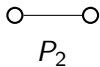
Questions of Interest

- Likelihood of k components?
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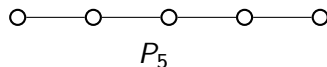
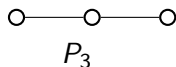
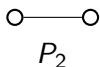
Questions of Interest

- Likelihood of k components?
- Determine k from π ?
- Structure of G changes k ?

Paths

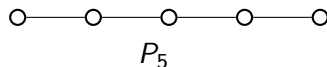
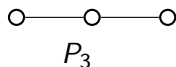
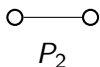


Paths



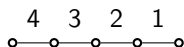
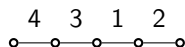
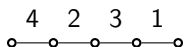
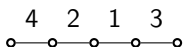
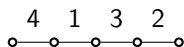
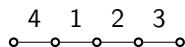
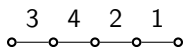
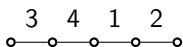
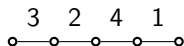
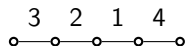
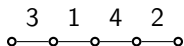
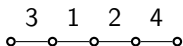
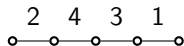
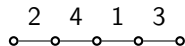
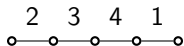
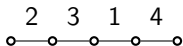
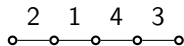
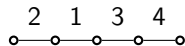
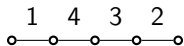
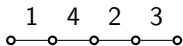
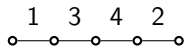
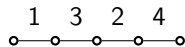
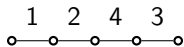
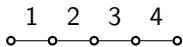
Path	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
P_2	1	-	-	-	-
P_3	2	-	-	-	-
P_4	4	2	-	-	-
P_5	8	16	-	-	-
P_6	16	88	16	-	-
P_7	32	416	272	-	-
P_8	64	1824	2880	272	-
P_9	128	7680	24576	7936	-
P_{10}	256	31616	185856	137216	7936

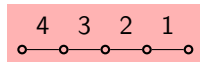
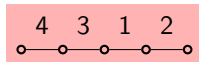
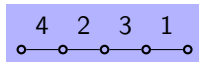
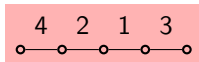
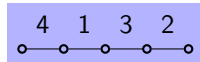
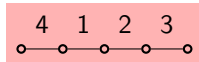
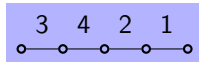
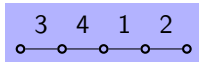
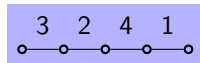
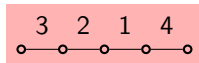
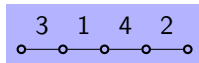
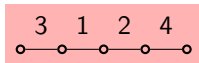
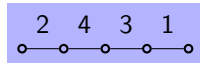
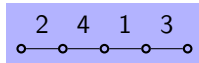
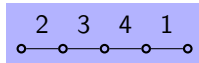
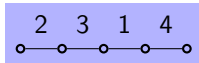
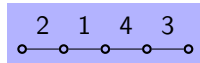
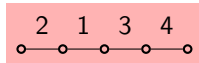
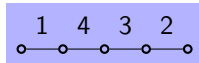
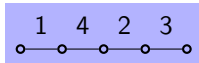
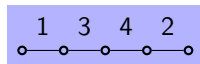
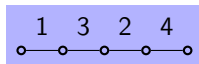
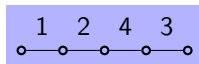
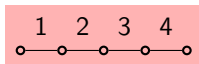
Paths



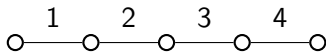
Path	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
P_2	1	-	-	-	-
P_3	2	-	-	-	-
P_4	4	2	-	-	-
P_5	8	16	-	-	-
P_6	16	88	16	-	-
P_7	32	416	272	-	-
P_8	64	1824	2880	272	-
P_9	128	7680	24576	7936	-
P_{10}	256	31616	185856	137216	7936



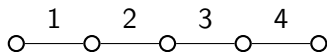


P_5 

Proof for P_5



Proof for P_5



1234

2134

2314

2341

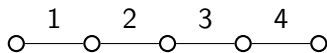
3214

3241

3421

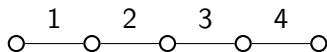
4321

Proof for P_5



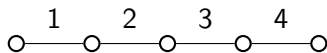
1234	}	1
2134		
2314	}	3
2341		
3214	}	3
3241		
3421	}	1
4321		

Proof for P_5



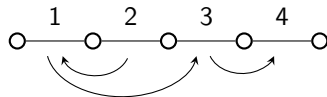
$$\begin{array}{lcl} 1234 & \} & 1 = \binom{3}{0} \\ 2134 & \} & \\ 2314 & \} & 3 = \binom{3}{1} \\ 2341 & \} & \\ 3214 & \} & \\ 3241 & \} & 3 = \binom{3}{2} \\ 3421 & \} & \\ 4321 & \} & 1 = \binom{3}{3} \end{array}$$

Proof for P_5

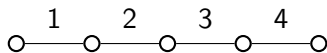


$$\begin{array}{lcl}
 1234 & \} & 1 = \binom{3}{0} \\
 2134 & \} & \\
 2314 & \} & 3 = \binom{3}{1} \\
 2341 & \} & \\
 3214 & \} & \\
 3241 & \} & 3 = \binom{3}{2} \\
 3421 & \} & \\
 4321 & \} & 1 = \binom{3}{3}
 \end{array}$$

(2134)

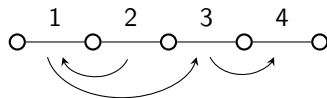


Proof for P_5

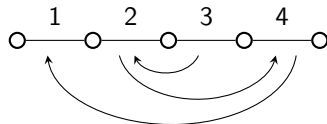


$$\begin{array}{lcl}
 1234 & \} & 1 = \binom{3}{0} \\
 2134 & \} & \\
 2314 & \} & 3 = \binom{3}{1} \\
 2341 & \} & \\
 3214 & \} & \\
 3241 & \} & 3 = \binom{3}{2} \\
 3421 & \} & \\
 4321 & \} & 1 = \binom{3}{3}
 \end{array}$$

(2134)



(3241)



1-Component Proof

Theorem

$$N(P_n, 1) = 2^{n-2}.$$

Proof.

P_n has size $m = n - 1$. The edges are labelled from left to right in ascending order. Let k be the first element of a permutation which builds one component, where $1 \leq k \leq m$.

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- 1.) Edges to the left of k must be considered in descending order, otherwise there is a gap. Similarly, edges to the right must be ascending.

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- 1.) Edges to the left of k must be considered in descending order, otherwise there is a gap. Similarly, edges to the right must be ascending.
- 2.) $k - 1$ left edges, $m - k$ right. Fixed orders, and edge k first:

$$\binom{m-1}{k-1} = \binom{n-2}{k-1}$$

1-Component Proof (cont.)

Theorem

$$N(P_n, 1) = 2^{n-2}.$$

Proof.

3.) Let k range $1 \leq k \leq m$, and combine:

$$\begin{aligned}\sum_{k=1}^{n-1} \binom{n-2}{k-1} &= \binom{n-2}{0} + \binom{n-2}{1} + \cdots + \binom{n-2}{n-2} \\ &= 2^{n-2}\end{aligned}$$



Binary Strings

1234

2134

2314

2341

3214

3241

3421

4321

Binary Strings

1234	→	1 + (RRR)
2134	→	2 + (LRR)
2314	→	2 + (RLR)
2341	→	2 + (RRL)
3214	→	3 + (LLR)
3241	→	3 + (LRL)
3421	→	3 + (RLL)
4321	→	4 + (LLL)

Binary Strings

1234	→	1 + (RRR)	→	000
2134	→	2 + (LRR)	→	100
2314	→	2 + (RLR)	→	010
2341	→	2 + (RRL)	→	001
3214	→	3 + (LLR)	→	110
3241	→	3 + (LRL)	→	101
3421	→	3 + (RLL)	→	011
4321	→	4 + (LLL)	→	111

Binary Strings

1234	→	1 + (RRR)	→	000
2134	→	2 + (LRR)	→	100
2314	→	2 + (RLR)	→	010
2341	→	2 + (RRL)	→	001
3214	→	3 + (LLR)	→	110
3241	→	3 + (LRL)	→	101
3421	→	3 + (RLL)	→	011
4321	→	4 + (LLL)	→	111

Corollary

There is a bijection φ between permutations which build 1 component on P_n and binary strings of length $n - 2$.

Binary Strings

1234	$\longrightarrow$	$1 + (\text{RRR})$	$\longrightarrow$	000
2134	$\longrightarrow$	$2 + (\text{LRR})$	$\longrightarrow$	100
2314	$\longrightarrow$	$2 + (\text{RLR})$	$\longrightarrow$	010
2341	$\longrightarrow$	$2 + (\text{RRL})$	$\longrightarrow$	001
3214	$\longrightarrow$	$3 + (\text{LLR})$	$\longrightarrow$	110
3241	$\longrightarrow$	$3 + (\text{LRL})$	$\longrightarrow$	101
3421	$\longrightarrow$	$3 + (\text{RLL})$	$\longrightarrow$	011
4321	$\longrightarrow$	$4 + (\text{LLL})$	$\longrightarrow$	111

Corollary

There is a bijection φ between permutations which build 1 component on P_n and binary strings of length $n - 2$.

Corollary

This can be extended to $\varphi(\alpha(\beta(\pi))) = \pi^{-1}$.

Paths

Path	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
P_2	1	-	-	-	-
P_3	2	-	-	-	-
P_4	4	2	-	-	-
P_5	8	16	-	-	-
P_6	16	88	16	-	-
P_7	32	416	272	-	-
P_8	64	1824	2880	272	-
P_9	128	7680	24576	7936	-
P_{10}	256	31616	185856	137216	7936

$$= 2^{n-2}$$

Paths

Path	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
P_2	1	-	-	-	-
P_3	2	-	-	-	-
P_4	4	2	-	-	-
P_5	8	16	-	-	-
P_6	16	88	16	-	-
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P_8	64	1824	2880	272	-
P_9	128	7680	24576	7936	-
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$$= 2^{n-2}$$

[A000431](#) Expansion of $2x^3/((1-2x)^2(1-4x))$.
(Formerly M2089 N0824)

+30
8

0, 0, 0, 2, 16, 88, 416, 1824, 7680, 31616, 128512, 518656, 2084864, 8361984, 33497088, 134094848, 536608768, 2146926592, 8588754944, 34357248000, 137433710592, 549744803840, 2199000186880, 8796044787712, 35184271425536, 140737278640128, 562949517213696 ([list](#); [graph](#); [refs](#); [listen](#); [history](#); [text](#); [internal format](#))

OFFSET 0,4

COMMENTS Number of permutations of length n with exactly one valley. Also (for $n > 0$), the number of ways to pick two of the 2^{n-1} vertices of an $n-1$ cube that are not connected by an edge. - [Aaron Meyerowitz](#), Apr 21 2014
 $a(n+1)$, $n \geq 1$: Number of independent vertex pairs for Q_n , $n \geq 1$: $2^{n-1} * (2^n - (n+1)) = T(2^n - 1) - n * 2^{n-1} = L_n - E_n = \text{A006516}(n) - \text{A001787}(n)$, where L_n is the number of vertex pairs and E_n is the number of vertex pairs yielding edges. (Cf. [A027624](#).) - [Daniel Forgues](#), Feb 19 2015

A000431	Expansion of $2x^3/((1-2x)^2(1-4x))$. (Formerly M2089 N0824)	+30 8
	0, 0, 0, 2, 16, 88, 416, 1824, 7680, 31616, 128512, 518656, 2084864, 8361984, 33497088, 134094848, 536608768, 2146926592, 8588754944, 34357248000, 137433710592, 549744803840, 2199000186880, 8796044787712, 35184271425536, 140737278640128, 562949517213696 (list ; graph ; refs ; listen ; history ; text ; internal format)	
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$$N(P_{n+1}, 2) = 2^{n-2}(2^{n-1} - n)$$

2-Component Proof

Theorem

$$N(P_{n+1}, 2) = 2^{n-2}(2^{n-1} - n).$$

Proof (sketch).

By induction.

- **Base Case:** $n = 3$. $\pi = 132$ or 312 , so expect 2.

$$2^{3-2}(2^{3-1} - 3) = 2^1(2^2 - 3) = 2(4 - 3) = 2\checkmark$$

- **Induction Hypothesis.** $N(P_{n+1}, 2) = 2^{n-2}(2^{n-1} - n)$

2-Component Proof

Proof (sketch).

- **Induction Step:** Consider $n + 1$ edges with $k = 2$.

2-Component Proof

Proof (sketch).

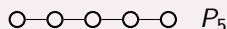
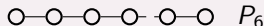
- **Induction Step:** Consider $n + 1$ edges with $k = 2$.
Remove last edge, shift down as needed. **e.g.** 12534 $\rightarrow$ 1243

2-Component Proof

Proof (sketch).

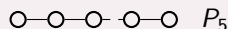
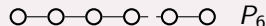
- **Induction Step:** Consider $n + 1$ edges with $k = 2$.
Remove last edge, shift down as needed. **e.g.** 12534 $\rightarrow$ 1243

$\pi_a : 12354$



$\psi_a : 1234$

$\pi_b : 12534$



$\psi_b : 1243$

- 1.) Close gap: Build 1 comp., add edge in middle ($n - 1$)
- 2.) Maintain gap: Build 2 comp., add edge on end (4)

2-Component Proof (cont.)

Theorem

$$N(P_{n+1}, 2) = 2^{n-2}(2^{n-1} - n).$$

Proof (sketch).

$$\begin{aligned} & 4 \cdot (2^{n-2}(2^{n-1} - n)) + (n - 1) \cdot 2^{n-1} \\ &= 2 \cdot (2^{n-1}(2^{n-1} - n)) + (n - 1) \cdot 2^{n-1} \\ &= 2^{n-1}(2^n - 2n) + (n - 1) \cdot 2^{n-1} \\ &= 2^{n-1}(2^n - 2n) + (n - 1) \\ &= 2^{n-1}(2^n - (n + 1)) \end{aligned}$$



Paths

Path	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
P_2	1	-	-	-	-
P_3	2	-	-	-	-
P_4	4	2	-	-	-
P_5	8	16	-	-	-
P_6	16	88	16	-	-
P_7	32	416	272	-	-
P_8	64	1824	2880	272	-
P_9	128	7680	24576	7936	-
P_{10}	256	31616	185856	137216	7936

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Path	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
P_2	1	-	-	-	-
P_3	2	-	-	-	-
P_4	4	2	-	-	-
P_5	8	16	-	-	-
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P_7	32	416	272	-	-
P_8	64	1824	2880	272	-
P_9	128	7680	24576	7936	-
P_{10}	256	31616	185856	137216	7936

General Recursion

Path	$k = 1$	$k = 2$	$k = 3$
P_2	1	-	-
P_3	2	-	-
P_4	4	2	-
P_5	8	16	-
P_6	16	88	16
P_7	32	416	272

Diagram illustrating the recursive calculation of P_k for $k=1$ to $k=7$. The values are shown in the table above. Red arrows labeled $\times 2$ indicate the progression from P_2 to P_3 , P_3 to P_4 , P_4 to P_5 , and P_5 to P_6 . Blue arrows labeled $\times 1$ indicate the progression from P_3 to P_4 and from P_5 to P_6 .

General Recursion

Path	$k = 1$	$k = 2$	$k = 3$
P_2	1	-	-
P_3	2	-	-
P_4	4	2	-
P_5	8	16	-
P_6	16	88	16
P_7	32	416	272

Diagram illustrating the recursive calculation of values for P_k across different paths (P_2 to P_7) and values of k ($k=1, 2, 3$). The values are shown in the table above.

Transitions between values are indicated by arrows:

- Red dashed arrows labeled $\times 0$ point from P_2 to P_3 , P_3 to P_4 , P_4 to P_5 , and P_5 to P_6 in the $k=1$ column.
- Red solid arrows labeled $\times 2$ point from P_2 to P_3 , P_3 to P_4 , P_4 to P_5 , and P_5 to P_6 in the $k=1$ column.
- Blue solid arrows labeled $\times 1$ point from P_3 to P_4 (in $k=2$) and from P_5 to P_6 (in $k=2$).
- Blue dashed arrows labeled $\times 0$ point from P_3 to P_4 (in $k=3$) and from P_5 to P_6 (in $k=3$).

General Recursion

Path	$k = 1$	$k = 2$	$k = 3$
P_2	1	-	-
P_3	2	-	-
P_4	4	2	-
P_5	8	16	-
P_6	16	88	16
P_7	32	416	272

General Recursion

Path	$k = 1$	$k = 2$	$k = 3$
P_2	1	-	-
P_3	2	-	-
P_4	4	2	-
P_5	8	16	-
P_6	16	88	16
P_7	32	416	272

Diagram illustrating the recursive construction of paths P_2 through P_7 for $k=1, 2, 3$. The paths are defined by the values in the table, with transitions indicated by arrows and multipliers.

- Red dashed arrows (labeled $\times 0$):** Indicate the downward sequence of values for each path.
- Red solid arrows (labeled $\times 2$):** Indicate the doubling of values from one path to the next.
- Blue solid arrows (labeled $\times 1$):** Indicate the addition of a new path.
- Green solid arrows (labeled $\times 2, \times 3, \times 4, \times 6$):** Indicate the branching of paths.
- Dashed blue arrows (labeled $\times 0$):** Indicate the termination of paths.

General Recursion

Theorem

$A(n, k) = N(P_{n+1}, k) =$ *Number of permutations on $[n]$ building k comp.*

$$A(n, k) = 2k \cdot A(n-1, k) + (n - 2(k-1)) \cdot A(n-1, k-1)$$

Proof (sketch).



General Recursion

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$A(n, k) = N(P_{n+1}, k) =$ *Number of permutations on $[n]$ building k comp.*

$$A(n, k) = 2k \cdot A(n-1, k) + (n - 2(k-1)) \cdot A(n-1, k-1)$$

Proof (sketch).

Build up to $[n]$ from $[n-1]$.



General Recursion

Theorem

$A(n, k) = N(P_{n+1}, k) = \text{Number of permutations on } [n] \text{ building } k \text{ comp.}$

$$A(n, k) = 2k \cdot A(n-1, k) + (n - 2(k-1)) \cdot A(n-1, k-1)$$

Proof (sketch).

Build up to $[n]$ from $[n-1]$.

1.) Maintain gap: Build k comp., add edge on end ($2k$)

$$2k \cdot A(n-1, k)$$



General Recursion

Theorem

$A(n, k) = N(P_{n+1}, k) =$ *Number of permutations on $[n]$ building k comp.*

$$A(n, k) = 2k \cdot A(n-1, k) + (n - 2(k-1)) \cdot A(n-1, k-1)$$

Proof (sketch).

Build up to $[n]$ from $[n-1]$.

1.) Maintain gap: Build k comp., add edge on end ($2k$)

$$2k \cdot A(n-1, k)$$

2.) Add gap: Build $k-1$ comp., add edge in middle ($n - 2(k-1)$)

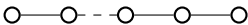
$$(n - 2(k-1)) \cdot A(n-1, k-1)$$



$$\left. \begin{array}{l} N(P_n, 1) = \pi \in [n-1] \text{ w/ 0 peaks} \\ N(P_n, 2) = \pi \in [n-1] \text{ w/ 1 peak} \\ \vdots \end{array} \right\} \text{Bijection?}$$

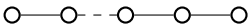
Peaks

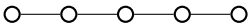
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Comps = 2

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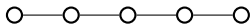
1324  Peaks = {3} ✓
Comps = 2

3421  Peaks = {4} ✗
Comps = 1

Inverses

$$\pi = 3421$$

$$\text{Comps} = 1$$

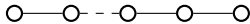


$$\pi^{-1} = 4312$$

$$\text{Peaks} = \emptyset$$

$$\pi = 4312$$

$$\text{Comps} = 2$$



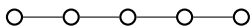
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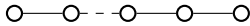


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$$\pi^{-1} = 3421$$

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π builds k comp. $\iff \pi^{-1}$ has $k - 1$ peaks

Proof for 2 Components (Revisited)

Theorem

The number of permutations on $[n]$ with 1 peak is:

$$\begin{aligned} N(P_{n+1}, 2) &= A(n, 2) = 2^{n-2}(2^{n-1} - n) \\ &= 4 \cdot (2^{n-3}(2^{n-2} - (n-1))) + (n-2) \cdot 2^{n-2} \end{aligned}$$

Proof (sketch).

To construct a perm. on $[n]$ from one on $[n-1]$, we can either:

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To construct a perm. on $[n]$ from one on $[n-1]$, we can either:

- 1.) Insert n onto an endpoint or next to an existing peak in a permutation with 1 peak. There are 4 insertion locations, and $2^{n-3}(2^{n-2} - (n-1))$ such permutations.

Proof for 2 Components (Revisited)

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The number of permutations on $[n]$ with 1 peak is:

$$4 \cdot (2^{n-3}(2^{n-2} - (n-1))) + (n-2) \cdot 2^{n-2}$$

Proof (sketch).

- 2.) Insert n into the middle of a permutation with no peaks, making a new peak. There are $n-2$ inner gaps between the $n-1$ existing numbers, and 2^{n-2} base permutations on $[n-1]$ without peaks



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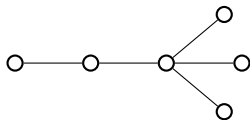
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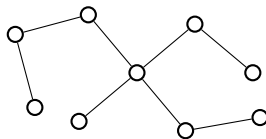
$$\Rightarrow 4 \cdot (2^{n-3}(2^{n-2} - (n-1))) + (n-2) \cdot 2^{n-2} \quad \checkmark$$



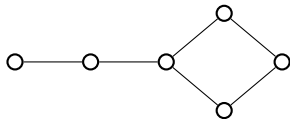
What else?



Brooms



Spiders



Lassos

What else?

$$N(B_{l,m}, 1) = \binom{l+m}{l} \cdot l! + \sum_{i=2}^{m-1} \sum_{j=i-2}^{m-2} \binom{j}{i-2} \binom{l+m-j-2}{l} \cdot l!$$

$$N(B_{l_1, l_2, m}, 1) = l_1! \cdot l_2! \cdot \left(\binom{l_1+l_2+m}{l_1} + \binom{l_1+l_2+m}{l_2} \right) \\ + \sum_{x=2}^{m-2} \binom{m-1+l_1+l_2}{x+l_1} \cdot l_1! \cdot l_2!$$

$$N(S_L, 1) = \sum_{j=1}^n \left[\left(\sum_{k=1}^n l_k - 1 \right) + \sum_{i=2}^{l_j} 2^{i-2} \left(\sum_{k=1}^n l_k - i \right) \right]$$

$$N(L_{n,m}, 1) = \sum_{x=0}^{n-1} \left(\binom{m+n}{m} \binom{n-1}{x} + \sum_{i=2}^m 2^{i-2} \binom{m+n-i}{m-i} \binom{n-1}{x} \right) \\ + \sum_{i=2}^x 2^{i-2} \binom{m+n-i}{m} \binom{n-1-i}{x-i} + \sum_{i=2}^{n-x-1} 2^{i-2} \binom{m+n-i}{m} \binom{n-1-i}{x} \right)$$

- 2-Components & Q_n

Future Work

- 2-Components & Q_n
- Other graph families
 $\hookrightarrow$ Computation limits

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- 2-Components & Q_n
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- General equations for $N(G, k)$

Thank you!

Questions?

